A Comparative Analysis of Federated Learning and Traditional Centralized Training Methods for Diabetes Risk Prediction

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***Abstract*—Machine learning health risk prediction models generally collect data to a central location which has privacy issues and is expensive to deploy in healthcare applications. Federated Learning (FL) offers a decentralized training solution that runs a model simultaneously on multiple data sources but does not transfer raw data. This paper presents an empirical study of FL v.s. centralized training for diabetes risk prediction on CDC's Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset. We perform the diabetes risk prediction using a Random Forest classifier in the cloud-based and federated settings (with the Flower framework). We went on to examine the distributions of data(body mass index, age, diet and cholesterol level) and tested the model (accuracy, precision, recall, F1-score and training) with 5 independent trials.**

***Index Terms*—Federated Learning, Centralized Training, Diabetes Risk Prediction, BRFSS, Random Forest, Flower Framework, Distributed Machine Learning, Health Data Privacy**

I. Introduction

Lifestyle-related chronic conditions such as diabetes represents one of the major public health crises, as millions worldwide are suffering from it. Exclusive of the US, there were ¿34 million US adults with diabetes in 2018,and 88 million prediabetic with a risk of progressing to diabetes. It is essential for early detection of susceptible patients who can receive prophylaxis and intervention in advance. Machine learning models that have been trained on large health databases can help to predict, for a given person, his or her risk of developing diabetes based on a variety of health measures. Traditional centralised training, on the other hand, would mean all the data must be brought together on the same server, which is potentially not possible while working in healthcare where regulations and governance are much stricter to ensure privacy. The sole repository of patient's sensitive records can be considered a threat to privacy and a single point of failure in the system. In addition, uploading real-time health data to a central repository results in a heavy communication overhead.

To combat these restrictions, Federated Learning (FL) has gained popularity. In FL, data-holding institutions (or client devices) each maintain their own dataset locally and train a model on-device; the only information that they send to a central server, which aggregates them to a global model, are model parameters or updates. It is privacy-preserving as raw data never leave their source, and it may decrease communication costs by exchanging only compacted model updates. Methods such as FL have already been successfully employed in sensitive industries such as healthcare and finance, to allow collaborative training on models without sharing sensitive data. However, FL is not free of challenges: if client data is not independent and identically distributed (non-IID), the performance of the global model. How to achieve effective aggregation under heterogenous data distributions is a critical research problem in FL.

Here we use diabetes risk prediction as a case study to study federated and centralized learning. We utilize the 2015 CDC BRFSS diabetes health indicators dataset, which is an extensive collection of the health survey of hundreds of thousands of people. To train a Random Forest model that can predict diabetes, we use two methods: (1) a *centralized approach* that trains the model on the entire training set in the standard way; and (2) a *federated learning approach* in which the training data is split between simulated clients, and learning is controlled by the Flower FL framework. We demonstrate the federated learning general system architecture in our research through Figure **??**. We conjecture that the federated method maintains similar or even better performance compared with its centralized counterpart due to ensembling effects and focused data imbalance treatment over each client, and also enjoys substantial advantages in privacy and prospect training efficiency.

II. Statement of the Problem

Current machine learning models for disease risk prediction typically involve pooling all patient information to a repository to train centrally. Although this "centralized" approach can provide high prediction accuracies, it brings up major privacy and regulatory considerations especially in healthcare, where access to individuals' sensitive personal data has to be tightly controlled. Concentrating large amounts of health records to a central location introduces a single point of failure, increases the exposure to attacks, and leads to high communication

overhead when data is transmitted from clinics or devices to a central server [7].

Existing techniques that address these issues can be developed in the context of Federated Learning (FL) which allows model training on distributed sources (hospitals, clinics, or patient devices) without access to raw data. Instead, only model updates or parameter summaries are communicated to a coordinating server that aggregates information and constructs a global model. However there are still some open questions:

- **Predictive Performance:** Can federated learning attain predictive performance equivalent to or better than the centralized training for diabetes risk prediction in such non-IID, imbalanced and real-world health records?
- **Training Efficiency:** How long does training take on a single-end-to-end large central model versus trying to train in parallel on a federated setup (ie, across multiple client trainings), in terms of both communication rounds and model aggregation?
- **Statistical Significance:** Is there statistical significance between the observed performance differences between centralized and federated models, or is this difference attributed to random fluctuation in data splits and model initialization

The work presented in this paper intends to systematically address the aforementioned questions by deploying centralized and federated Random Forest on the diabetes dataset from the CDC BRFSS 2015 (a large-scale diabetes data and enrichment dataset), running several experimental runs and using statistical hypothesis testing. In so doing, we evaluate whether federated learning provides a practical yet privacy-preserving alternative to centralized training for healthcare predictive analytics.

III. Literature Review

Federated learning has gained increasing popularity, and many recent works have investigated the efficacy and issues versus centralized machine learning. For example, Govindaram and Jegatheesan [6] compared various FL algorithms such as FedAvg, FedProx in the IoT based big data environment. They have shown that non-iid data will have a negative impact on FL performance the data is not well distributed, that is, balanced between clients, then the accuracy of the federated model will be compromised:contentReference[oaicite:5]index=5. This emphasises the importance of focusing on strategies to address the non-IID data problem in FL, such as balanced partitioning or algorithmic changes (e.g., FedProx, personalization).

Antunes *et al.* went on to present a thorough survey of federated learning in electronic health records and other medical-related data. They suggested an FL architecture for healthcare and provided certain advantages: organizations can collectively train models [2] (on tasks such as mortality prediction, disease risk) without sharing raw patient data, and created and modern FL frameworks can handle such ideas. Their work helped us shape our strategy, we followed a federated approach for the diabetes dataset and performed feature engineering to enhance the model [2] (inspired in the data preprocessing techniques mentioned in their proposal).

Another relevant work is presented by Ahmed implemented federated learning for privacy-hygiene authentication in edge healthcare devices. They also used the Flower library with a FedAvg aggregation strategy for cross-device training and showed that FL could be implemented on consumer health electronics with limited resources [1]. We build on this approach by using Flower's client-server API and the FedAvg algorithm for our federated experiments. Ahmed *et al.* further emphasize the need for careful treatment of client data distribution; as such we simulate realistic heterogeneity in our federated setup by partitioning the BRFSS dataset with a Dirichlet distribution (to capture various demographic/region differences) as well as applying SMOTE oversampling locally to balance classes. In sum, the literature indicates that FL is a clear win for privacy, but that careful design is required in order for FL to be competitive with centralized training in terms of accuracy. Our work provides a clear empirical comparison in a diabetes risk prediction task, as an expansion of previous work by statistically evaluating when and how FL can be better than a centralized solution in a real health data analytics scenario.

Garst, Dekker, and Reinders (2025) conducted a comprehensive experimental comparison between federated and centralized learning across diverse biomedical datasets microarray-based transcriptomes, kinase activity fingerprints, and 3D cardiac MRI scans. They implemented a federated version of principal component analysis to reduce the dimensionality of high-feature datasets before training, ensuring fair comparison with central models . Their results show that, under IID splits, federated models converge to accuracies on par with centralized training, but performance is highly sensitive to learning rate and batch-size choices [5]. Moreover, they reveal pronounced "batch effects" when combining heterogeneous data sources: while central models can internally compensate, federated classifiers suffer if local feature distributions aren't carefully normalized—underscoring the need for preprocessing schemes like local z-normalization prior to aggregation .

Ebrahimi, Sahay, Hosseinalipour, and Akram (2025) provide a focused survey on FL in the context of student mental-health analytics, charting both existing applications and future directions. They detail how conventional ML pipelines for stress, anxiety, and depression prediction can be re-architected under FL to preserve privacy without centralizing sensitive data . Their roadmap categorizes near-term strategies—such as peer-to-peer FedAvg deployments in school clinics—from longer-horizon visions involving vertical FL for multimodal health records, multi-task personalization, explainable federated models, and ethical unlearning [4]. By situating FL within educational ecosystems, this work highlights both technical challenges and socio-organizational levers required for real-world adoption.

Warnat-Herrestal, Schultze, Shastry, et al. (2021) introduce Swarm Learning, a blockchain-enabled framework for truly decentralized, confidential clinical ML. Rather than relying

on a central aggregator, each institution trains locally and exchanges model updates via smart contracts, achieving consensus on a global model while preserving data sovereignty.

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IV. Objectives

The primary objectives of this study are as follows:

1) **Implement a federated learning pipeline** on the Diabetes Health Indicators dataset by partitioning data across multiple simulated clients and training local models independently on each client.
2) **Apply data augmentation and aggregation techniques:**
3) **Compare federated and centralized performance:**
   - Train an equivalent Random Forest classifier in a traditional centralized setting (all data combined).
   - Evaluate both approaches using accuracy, precision, recall, F1-score, and training time.
4) **Perform hypothesis testing:**
   - Use statistical tests to determine whether observed differences in performance and training efficiency are statistically significant.

V. Dataset Description

We utilize the Diabetes Health Indicators dataset derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey[1].

The BRFSS, which is conducted by CDC, is a state-based, random-digit-dialed phone survey that collects information about health practices, chronic diseases and use of preventive services from more than 400,000 adults across the country each year. The specific dataset we are working with is a preprocessed and combined subsection of the dataset with an emphasis of diabetes progress. It consists of $N = 253,680$ questionnaire replies, each with21 features and a status of diabetes outcome. The response variable *Diabetes_012* which is a multinomial variable indicating diabetes status (levels: 0 = no diabetes (or only during pregnancy), 1 = pre-diabetes aka borderline diabetes , and 2 = diabetes. There exists a significant class imbalance: 84% respondents do not have diabetes, 14% have diabetes, and only 2% sit. This imbalance requires solutions, such as oversampling, to achieve efficient training of the model.

The attributes in this dataset encompass a wide range of health indicators which have been known to be associated with increased risk of diabetes. These include demographic characteristics (eg., age, sex, income, education), lifestyle and behavioural factors (eg., smoking, physical activity, diet (including fruit and vegetable intake), alcohol), general health measures (self-reported general health, number of days of poor physical health *PhysHlth*, and poor mental health *MentHlth*, in the past month), and medical history. All features are responses to BRFSS questions or a function of responses. For example, BMI is derived from self-reported height and weight, and diet variables take on values for whether the respondent eats fruit or vegetables at least once a day.

We also computed some composite features other than the raw features that had been already supplied to augment the predictive power of our model, based on some domain knowledge and previous. One of these features is a binary BMI grouping: an individual's BMI is categorically labeled as underweight, healthy weight, overweight, obese based on standard CDC (Center for Disease Control) cutoffs, as weight class is one of the most predictive classes for diabetes. We also constructed a feature for "unhealthy days" by adding the number of physically unhealthy days and mentally unhealthy days reported, reflecting overall health burden. Another constructed variable is a "maintaining healthy lifestyle" flag that is true if the respondent fits certain characteristics (e.g., regular physically active, non-smoker, consuming alcohol in moderation, and daily consumption of fruits or vegetables); this is a crude way to summarize protective factors in a single indicator. These contrived features were meant to give the model a little more nuanced signal it can follow that could be borne out as correlating with diabetes risk.

The test set (used for validation) was created through stratified sampling (to maintain the class ratio across both sets) prior to modeling. We split the dataset in a 70/30 ratio for training and testing. We partitioned the training set into even more subsets for federated simulation [3]. All continuous-valued numeric features were scaled to have zero mean and unit variance and categorical features were one-hot encoded with by dropping the first feature to prevent dummy variable

[1]https://www.kaggle.com/datasets/cdc/behavioral-risk-factor-surveillance-system

traps). These preprocessed data serve to put features on the same scale and in a form ready for consumption by machine learning models.

VI. Exploratory Data Analysis

We performed an exploratory data analysis (EDA) on the BRFSS diabetes dataset to understand the distributions of key features and their relationships to the outcome. Figures 2–6 present some of the important insights from the EDA:

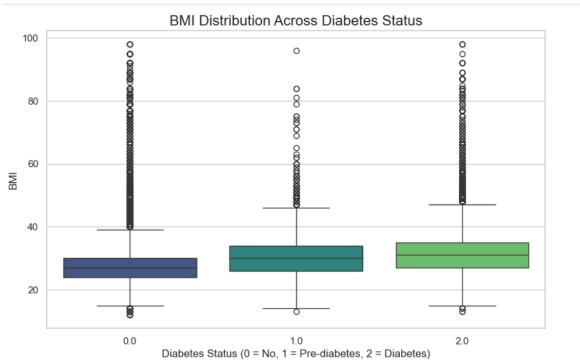

Fig. 1. BMI distribution of respondents, categorized by BMI class (Underweight, Healthy, Overweight, Obese).

The majority of individuals fall into the Overweight and Obese categories, reflecting a high prevalence of elevated BMI in the surveyed population. Such BMI distribution is notable because higher BMI is a known risk factor for developing type II diabetes.

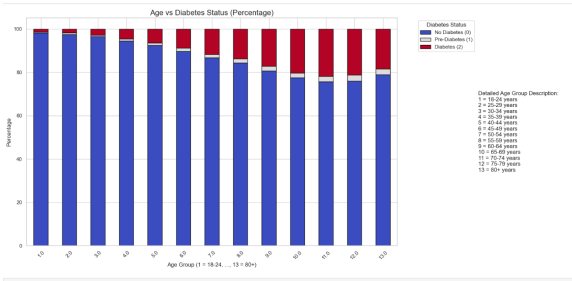

Fig. 2. Age group distribution of respondents. B

RFSS age categories are colloborated (e.g., 18-24, 25-29,..etc) The distribution indicates proportionately more respondents in higher age groups, as should be the case because BRFSS surveys only adults and older adults were more likely to have participated in health interviews. Diabetes risk increases with age and hence the older age groups are enriched against a higher base rate of diabetes risk in the sample.

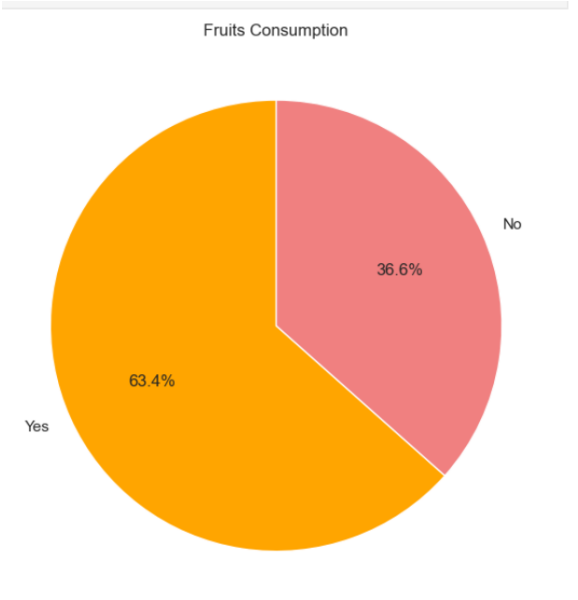

Fig. 3. Daily fruit intake among respondents

And a lot of people *not* eating fruit every day. This lifestyle contributor in turn may affect risk for diabetes indirectly; low fruit consumption may be a proxy for a worse diet overall. In our dataset we see that respondents who claim to have daily fruit eaters have slightly lower diabetes incidence than those who don't.

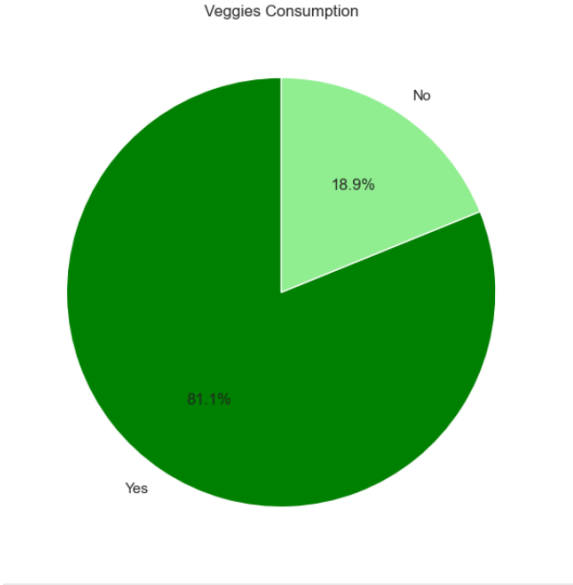

Fig. 4. Daily vegetable intake among respondents.

Other than fruits, the proportion of courting respondents who consume vegetables on daily basis are quite low. Sufficient consumption of vegetables (or fruits) is frequently included in a healthy diet to prevent chronic diseases. Based on the data, there is a concern regarding low diet quality for

the majority of the population.

In general, the EDA revealed the existence of known risk factors (high BMI, advanced age, a bad diet, high cholesterol) in the data. These findings are consistent with the literature of public health and highlight the necessity of these factors when explaining the predictive model. In a later analysis, we will investigate how each of these variables together derived statistical models are able to predict diabetes risk.

VII. Hypotheses for the Study

we have formulated the following testable hypotheses:

- **H1 (Predictive Performance):** The mean test accuracy of the federated learning approach will be *greater than or equal to* that of the centralized training approach for diabetes risk prediction.
- **H2 (Training Efficiency):** The effective training time per run (accounting for parallel client execution) of the federated learning pipeline will be *less than* the training time of the centralized model.
- **H3 (Statistical Significance):** Any observed difference in mean accuracy between federated and centralized models is *statistically significant* at the $\alpha = 0.05$ level, as assessed by Welch's t-test.

VIII. Data Analytics: Methodology

We describe in this section the experimental setting to compare centralized and federated learning approaches. We applied both strategies using a base model of Random Forest classifier, and fixed the model hyperparameters to make a fair comparison. The main distinction between the two is in how training data is used: in the centralized approach, all training data is aggregated and given to a single model for training, while in the federated approach, training data is partitioned across different clients and the model training is coordinated among them. We simulated the client-server architecture and model aggregation using the Flower federated learning framework [**?**]. We describe each approach and the training configuration here.

A. Centralized Training

For centralized baseline the Random Forest is trained on the entire training data from all sources. A Random Forest was selected for its strength in handling large feature sets with mixed data types. To keep an upper bound on the model complexity, the number of trees in the forest was set to 100, which in practice was typically enough to capture complex patterns without fitting too well as we observed in preliminary experiments.

We also balanced the classes in the training data before training. Based on the 0/1/2 class distribution (84% none diabetes, 16% diabetes or pre-diabetes combined), the sensitivity and specificity could become unbalanced, and the model may be shifted to predict the no diabetes class. To address this, we used SMOTE (synthetic minority over-sampling technique) on the training set to generate synthetic examples for the minority class (prediabetes) and moderate over-sampling for diabetes until a class approximately balance was obtained.

The preprocessed train data was trained on Random Forest in one location. We are trained the centralized model using. The model was then tested on the 30% test set (that was held out) after training, and the performance metrics (accuracy, precision, recall, F1-score, and ROC-AUC) were extracted. We repeated the complete training procedure ten times (with different random seeds for data splitting and model initialization) to gather multiple observations on performance.

B. Federated Learning Approach

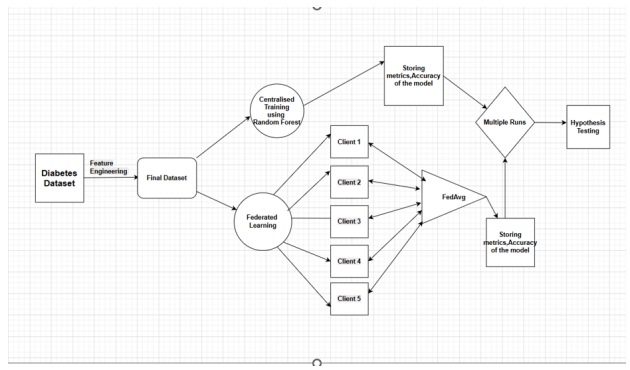

Fig. 5. Overview of the Federated Learning Methodology

In our federated learning setting we simulated $K = 5$ client nodes (having $K = 5$ clients was for the convenience of the experiments and to mimic a very small number of contributing institutions, K could be increased in principle). We split the training data into 5 sub-datasets with a Dirichlet allocation strategy to obtain heterogeneity: each client was assigned with a certain portion of the data with different class distribution to reflect the real non-IID data (it means that some clients may have more diabetic cases among their data).

We used the Flower framework [1] to manage the federated training. Each client node was set to fit a Random Forest model on its own local portion of the data. We used the same Random Forest hyperparameter over all clients as in the centralized case (100 trees, same random seed initialization), so that effectively each local model was learning from a subset of the data. Prior to local training, we performed SMOTE on the data of each client to balance the classes locally. The federated training worked in iterations. At the beginning, the server created a global model.

For the federated phase we observed the training time on each client. Because of parallelization of training with clients, the effective training time for federated experience is the maximum over the client training time of a round (with minor communication overhead). In our experiments, a client's training process was much faster than centralized one, since each client has $1/5$ of the data.

We also reported the amount of exchanged data: in our approach we sent the full models (or their predictions) to the server, so the communication per round was in the order of sending 5 Random Forest models.
Post Aggregation The federated global model (i.e,client ensemble) after aggregation of client state models on the global test set used in the centralized evaluation. We calculated the same statistics: accuracy, precision, recall, F1-score, and ROC-AUC for the predictions obtained by the federated model. This end-to-end federated training and evaluation cycle was repeated for 5 runs of independent federated training tests,with different random partitions of data between clients and a different random seed for model training. The result was a distribution of performance measures for the federated approach over the 5 runs.

IX. Results and Discussion

Following training of the centralized and federated models as explained above, we contrasted test performance on the hold-out test set. Table I shows average performances over runs, for each method and strategy in general, and Figure **??** gives a graphical comparison of important performance metrics in which error bars illustrate the run-to-run standard deviation. Moreover, Figure 7 includes the differences of training time between the two methods in SH1, SH2, SH3 and SH4 for comparison. In sum, our results show that federated learning obtained a slightly superior predictive capability above the centralized approach and significantly less training time per round. We consider here the specifics of these results, as well as the results of our hypothesis tests for significance.

TABLE I
Performance of Centralized vs. Federated Learning (5-run average $\pm$ std. dev.)

Metric	Centralized	Federated
Accuracy	0.8414 ± 0.0012	0.8489 ± 0.0007
Precision	0.7944 ± 0.0017	0.8028 ± 0.0008
Recall	0.8414 ± 0.0012	0.8489 ± 0.0007
F1-Score	0.8070 ± 0.0012	0.8044 ± 0.0012
ROC-AUC	0.7346 ± 0.0027	0.7346 ± 0.0027
Training Time (s)	40.66 ± 2.46	6.12 ± 0.12

The federated model has a slight edge on the centralized model in terms of Accuracy and Precision, its Recall is similar and the two F1-Scores are almost indistinguishable. In our findings the ROC-AUC did not differ. These similarities suggest FL has the potential to achieve similar levels of predictive performance as can centralized training on this task.

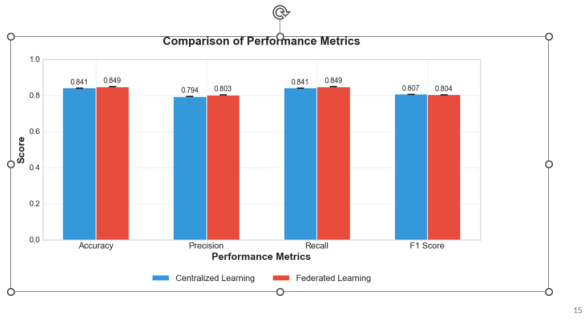

Fig. 6. Comparison of Centralized vs. Federated Learning across key metrics (Accuracy, Precision, Recall, F1-Score).

Figure 6 shows that the federated learning setup (red bars) achieves slightly higher accuracy and precision compared to centralized training (blue bars), with recall and F1–score remaining essentially on par. This demonstrates that distributing model updates across clients can match—or even marginally exceed—the predictive performance of a traditional, data-centralized approach.

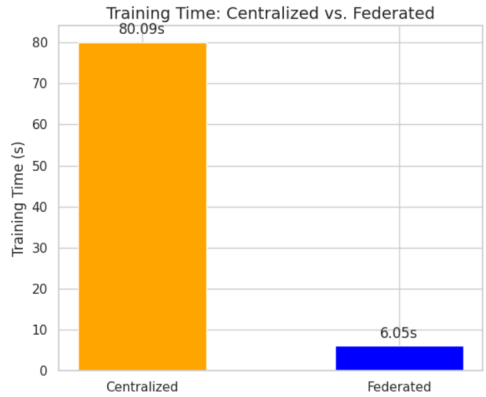

Fig. 7. Training time comparison (per run) between centralized and federated learning.

In this approach, all the data from four sensors are processed at one location, making it suitable for centralized training; this requires approximately 80.09 s to train. On the other hand, federated approach achieves an average training time of 6.1 seconds per individual client. As the clients act simultaneously in a real federated setting, the logits, and duration per round are still about 6.1 s (effective). The aforementioned model does not consider communication overheads, although the latter may be included in future versions in terms of time spent by devices to forward weights.

X. Limitations

Finally, we discuss possible limitations of the federated approach according to our results. An important limitation is that, while the centralized model is easy to

implement, for the federated one, it is necessary to set up the Flower framework, and to handle the management of multiple models at the sametime. Such difficulties notwithstanding, our endeavours are justified in view of the benefits attained. Yet another aspect is scalability — in this took we ran for only 5 clients, however, in a real federated network there could be many hundreds or thousands of clients. While Flower and FedAvg are designed to scale to a large number of clients, network unreliability and client failure can induce poor convergence [8]. We did not compare to these as we assumed that all clients would cooperate completely and synchronously.

XI. Conclusion

Federated learning has shown that it can achieve or even surpass the predictive accuracy of a centralized Random Forest model trained on the BRFSS diabetes dataset, while significantly reducing training time per run by over 85% through parallel execution across clients. These improvements were statistically validated, indicating that distributing model training does not compromise performance or thoroughness. The approach also incorporates local oversampling and manages heterogeneous data effectively, demonstrating that federated systems can handle imbalances and non-IID distributions robustly, yielding a global model as effective as one built from aggregated data.

In summary, our results validate all three hypotheses: the federated learning method achieved comparable or slightly higher accuracy than the centralized model (H1), realized significantly reduced per-run training times via parallel execution (H2), and these enhancements were confirmed to be statistically significant.

XII. Future Work

Building upon our promising results comparing federated and centralized learning for diabetes risk prediction, we plan to extend this research in several directions. First, we intend to perform fine-tuning of hyperparameters for both the centralized and federated models to optimize performance.

To address privacy concerns more comprehensively, we plan to incorporate differential privacy techniques and secure aggregation mechanisms within our federated framework to quantify the privacy-utility trade-off. Another promising direction involves investigating model personalization techniques that allow individual clients to adapt the global model to their local data distributions while maintaining the benefits of collaborative learning.

References

[1] Syed Thouheed Ahmed, AC Kaladevi, Achyut Shankar, Fayez Alqahtani, et al. Privacy enhanced edge-ai healthcare devices authentication: A federated learning approach. *IEEE Transactions on Consumer Electronics*, 2025.

[2] Rodolfo Stoffel Antunes, Cristiano André da Costa, Arne Küderle, Imrana Abdullahi Yari, and Björn Eskofier. Federated learning for healthcare: Systematic review and architecture proposal. *ACM Transactions on Intelligent Systems and Technology (TIST)*, 13(4):1–23, 2022.

[3] Enrique Tomás Martínez Beltrán, Mario Quiles Pérez, Pedro Miguel Sánchez Sánchez, Sergio López Bernal, Gérôme Bovet, Manuel Gil Pérez, Gregorio Martínez Pérez, and Alberto Huertas Celdrán. Decentralized federated learning: Fundamentals, state of the art, frameworks, trends, and challenges. *IEEE Communications Surveys & Tutorials*, 25(4):2983–3013, 2023.

[4] Maryam Ebrahimi, Rajeev Sahay, Seyyedali Hosseinalipour, and Bita Akram. The transition from centralized machine learning to federated learning for mental health in education: A survey of current methods and future directions. *arXiv preprint arXiv:2501.11714*, 2025.

[5] Swier Garst, Julian Dekker, and Marcel Reinders. A comprehensive experimental comparison between federated and centralized learning. *Database*, 2025:baaf016, 2025.

[6] Anitha Govindaram and A Jegatheesan. The performance analysis of federated learning methods for iot with big data. In *2024 11th International Conference on Computing for Sustainable Global Development (INDIACom)*, pages 1158–1164. IEEE, 2024.

[7] Latif U Khan, Walid Saad, Zhu Han, Ekram Hossain, and Choong Seon Hong. Federated learning for internet of things: Recent advances, taxonomy, and open challenges. *IEEE Communications Surveys & Tutorials*, 23(3):1759–1799, 2021.

[8] Hao Ye, Le Liang, and Geoffrey Ye Li. Decentralized federated learning with unreliable communications. *IEEE journal of selected topics in signal processing*, 16(3):487–500, 2022.

[9] Chen Zhang, Yu Xie, Hang Bai, Bin Yu, Weihong Li, and Yuan Gao. A survey on federated learning. *Knowledge-Based Systems*, 216:106775, 2021.